brilliant minds

Cognitive Load Reduction Assistant

Microsoft Innovation Challenge March 2026

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BRILLIANT MINDS

- **Brilliant Minds is** a Cognitive Load Reduction Assistant.
- **AI-powered learning initiative.**
- **Designed to support neurodiverse learners to build cognitive resilience.**
- **Promotes inclusion and equal opportunities in education.**
- **Focused on advancing STEAM subjects (Science, Technology, Engineering, Arts, Mathematics).**
- **Supports collaboration between students, teachers, and parents.**
- **Vision: To empower the next generation to learn anytime, anywhere, without limits.**
- **Mission: Use Responsible AI and Microsoft technologies to provide personalized, inclusive, and flexible learning experiences.**

- **Microsoft Azure AI Foundry**
- **Microsoft Agent Framework**
- **Microsoft Azure Document Intelligence**
- **Microsoft Azure Cosmos DB**
- **Microsoft Azure Blob Storage**
- **Microsoft Azure AI Search**
- **Large Language Models**
- **Compliance with Responsible AI principles**
- **Python.**
- **FastAPI.**
- **Nextjs**

PROBLEM DESCRIPTION

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Autism Profile

- Hyper-focused, deep, and rigid attention.
- Difficulty comprehending implicit social rules.
- Prefers strict routines and absolute stability.

ADHD Profile

- Dispersed, unstable, but capable of extreme hyper-focus on novel stimuli.
- Impulsive, fast-paced processing.
- Constantly seeks novelty and stimulation.

Dyslexia Profile

- High spatial and creative intelligence.
- Phonological processing (manipulating sounds).
- Laborious decoding of written text.

DIAGNOSTIC OVERLAP: Approximately 50% of autistic individuals also present with ADHD. The AI architecture must adapt to overlapping, dual-profile cognitive demands simultaneously.

PROBLEM DESCRIPTION

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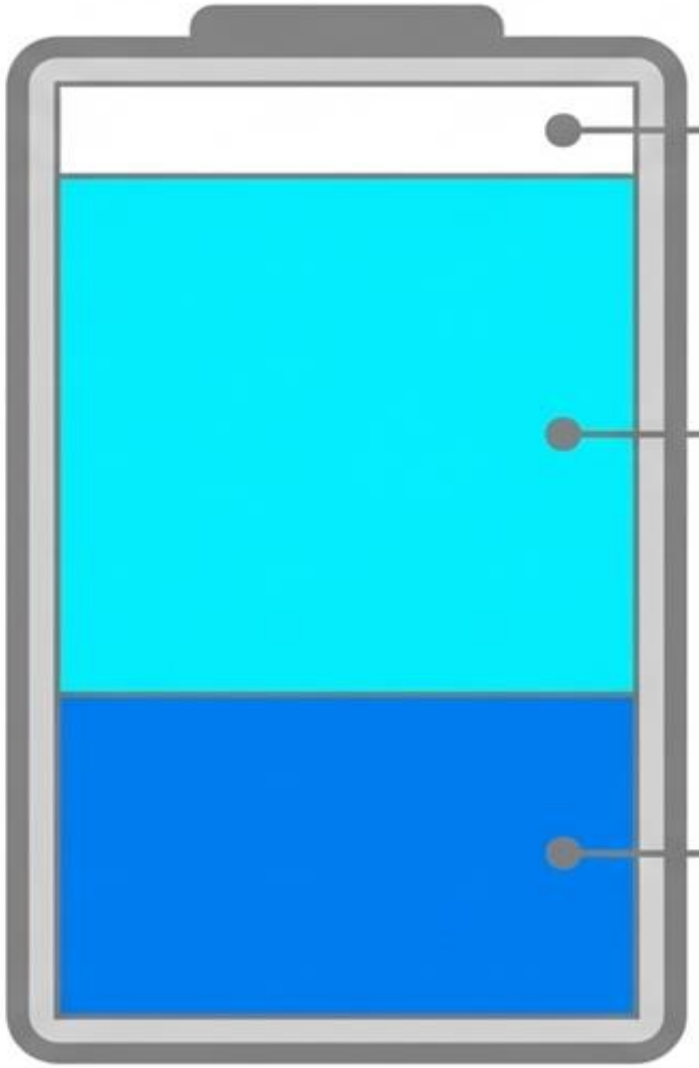
Cognitive Load Theory. A brain overwhelmed by excessive Extrinsic Load has zero capacity left for Germane Load. The student survives the environment but learns nothing.

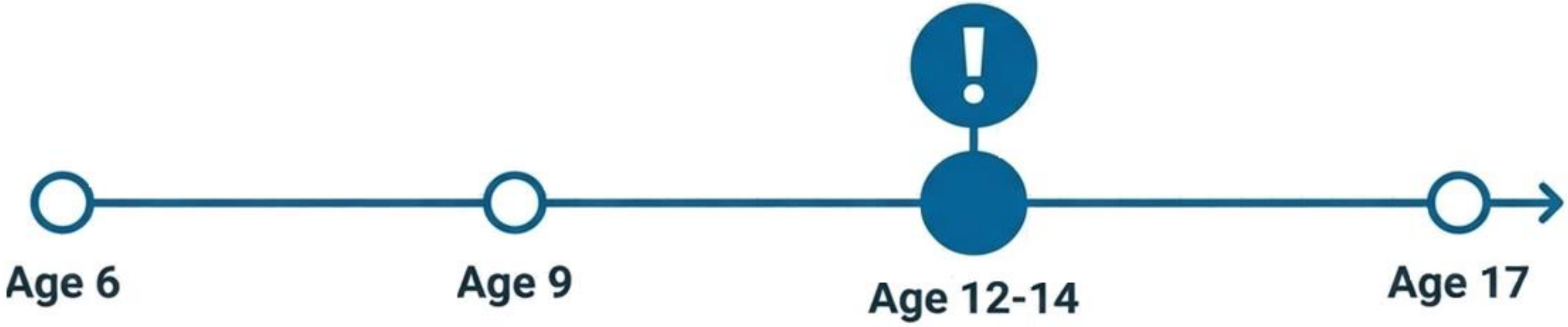


GERMANE LOAD: The mental space required to actually learn, connect ideas, and store information. (Currently zero capacity).

EXTRINSIC LOAD: Bad formatting, noisy environments, ambiguous instructions. Must be minimized.

INTRINSIC LOAD: The natural difficulty of the STEAM concept (e.g., a physics equation). Cannot be reduced.

PROBLEM DESCRIPTION



The Scaffolding Collapse

Middle school marks the exact developmental phase where external organizational support from parents and elementary teachers is abruptly withdrawn.

The Executive Demand

Students are suddenly expected to manage their own time, prioritize complex tasks, and self-regulate without a fully developed executive function framework.

The Intervention Point

This is the moment neurodivergent students are statistically most vulnerable to abandoning STEAM subjects. The AI bridges this fatal developmental gap.

PROBLEM DESCRIPTION

Time Limits: Triggers ADHD time blindness and Autistic chronophobia.

Dense Text Blocks: Overloads Dyslexic phonological processing.

Ambiguous Instructions: Triggers Autistic literal processing confusion.

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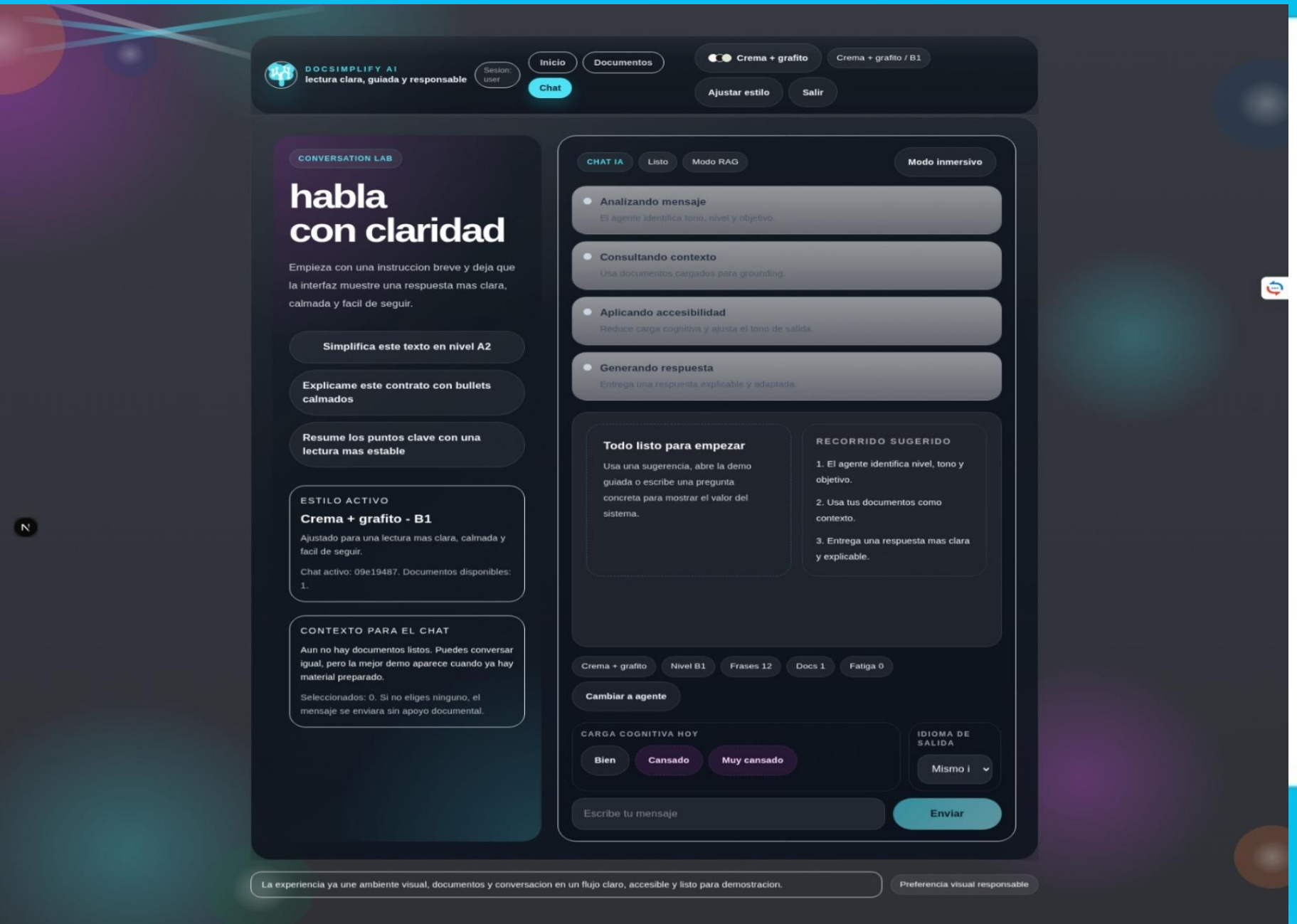
USER INTERFACE FUNCIONALITY



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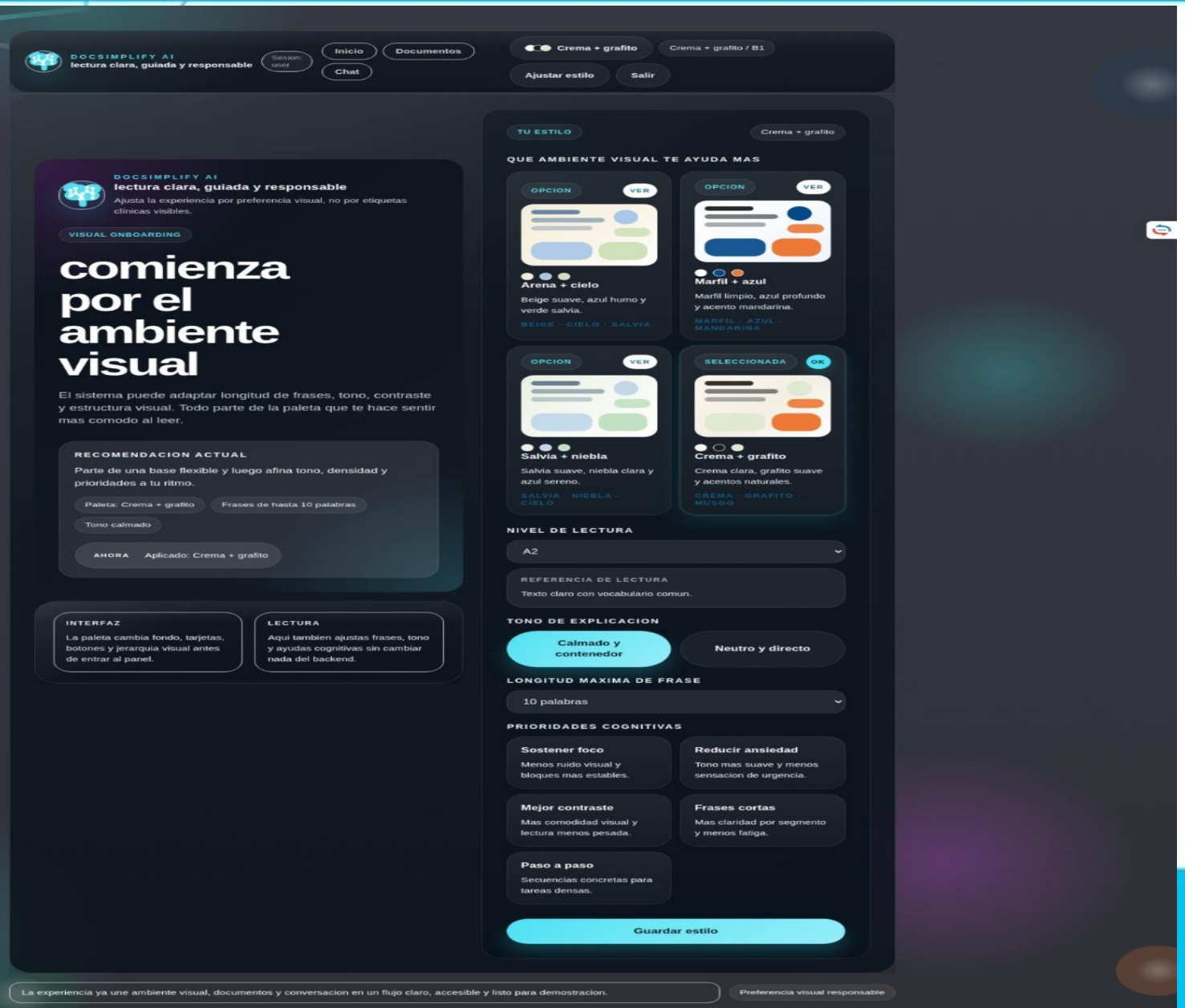

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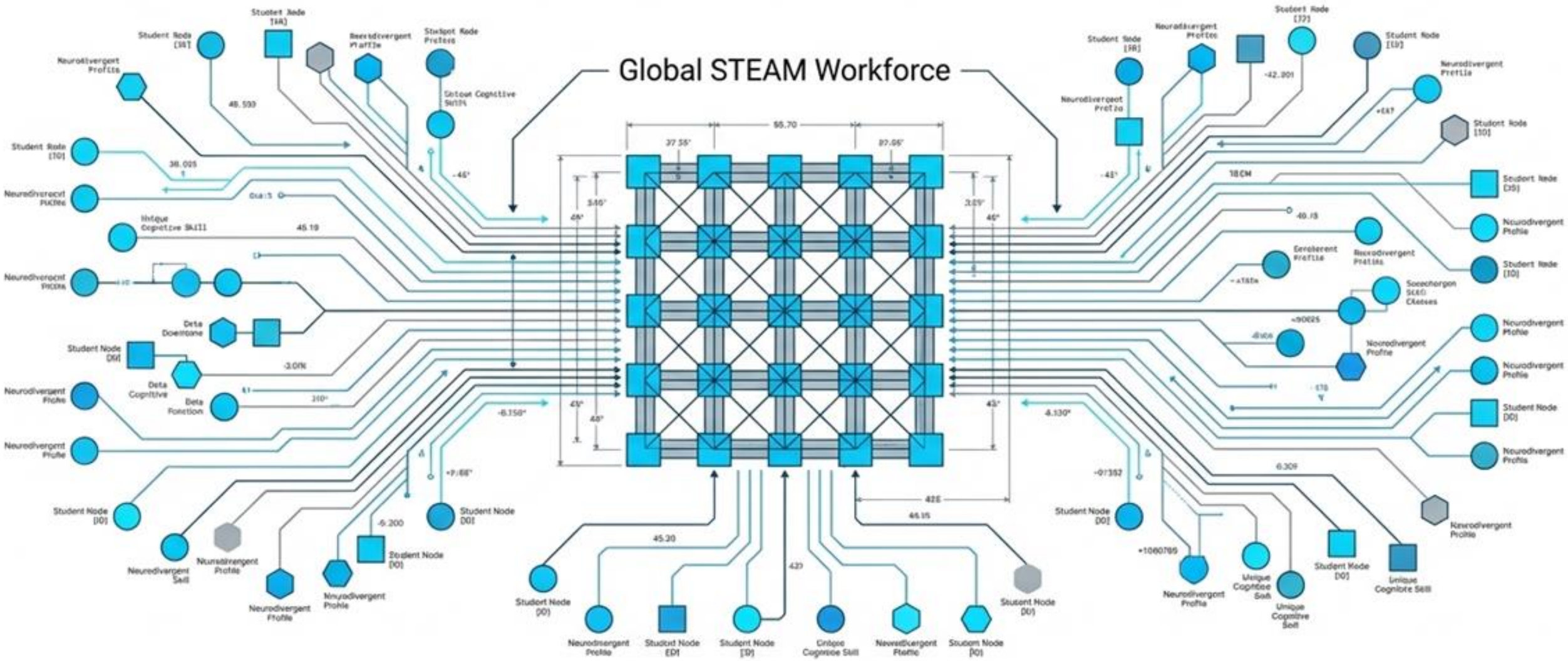
Global STEAM Workforce

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